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Visual Analytics and Comparative Evaluation of Machine Learning Models for Heart Disease Prediction

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ABSTRACT:

Cardiovascular diseases remain one of the leading causes of mortality worldwide, underscoring the urgent need for accurate and efficient diagnostic tools. This study presents a comparative evaluation of classical machine learning models for heart disease prediction using the UCI Heart Disease dataset. Comprehensive preprocessing steps, including missing value handling, outlier removal, normalization, and categorical encoding, were applied to ensure data quality. Four classifiers—Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), and Support Vector Machine (SVM)—were trained and validated using 10-fold cross-validation. Performance was assessed using accuracy, precision, recall, F1-score, and ROC-AUC. Experimental results demonstrate that Random Forest achieved the highest accuracy (88%) and ROC-AUC (0.91), outperforming other models, while Logistic Regression provided interpretable coefficients suitable for clinical contexts. Visualization techniques such as bar charts, ROC curves, and feature importance plots were employed to enhance interpretability and support clinical trust. The findings highlight the balance between predictive performance and explainability, offering a reproducible benchmark for future research. Future work will explore hybrid feature selection methods, ensemble frameworks integrating deep learning, and validation on real-world clinical datasets to improve generalizability and clinical adoption.

INDEX TERMS:

Cardiovascular disease, heart disease prediction, machine learning, logistic regression, decision tree, random forest, support vector machine, feature selection, ensemble learning, data preprocessing, visual analytics, ROC-AUC, interpretability.

I. INTRODUCTION

Cardiovascular diseases are one of the leading causes of death worldwide [1]. These conditions encompass a wide range of disorders affecting the heart and blood vessels, including coronary artery disease, arrhythmias, congenital heart defects, and heart failure. Accurate and timely diagnosis of heart disease is critical, as early intervention can significantly reduce mortality and improve patient outcomes [2]. Traditional diagnostic methods rely heavily on clinical expertise,

Physical examinations, and patient history, which, while valuable, are often limited by subjectivity and variability in interpretation [3].

In recent years, the integration of machine learning (ML) into healthcare has opened new avenues for predictive modeling and decision support. ML algorithms can process large volumes of patient data, uncover hidden patterns, and generate predictive insights that aid clinicians in identifying at-risk individuals [4]. The UCI Heart Disease dataset, a widely used benchmark in medical ML research, provides a structured representation of patient attributes such as age, cholesterol levels, blood pressure, and ECG results, making it suitable for developing and evaluating classification models [5].

Despite the proliferation of advanced deep learning architectures, classical ML models such as Logistic Regression (LR), Decision Trees (DT), Random Forests (RF), and Support Vector Machines (SVM) remain popular due to their interpretability, ease of implementation, and relatively low computational requirements [6]. Moreover, visualizing model outputs and feature relationships enhances transparency and supports clinical trust, especially in settings where explainability is as important as accuracy [7].

This study aims to perform a comparative analysis of standard ML classifiers on the UCI Heart Disease dataset, emphasizing performance metrics such as accuracy, precision, recall, and F1-score. Additionally, we employ visualization techniques to explore feature correlations, model decision boundaries, and classification outcomes. By establishing clear baselines and interpretive visuals, this work contributes to the development of accessible and trustworthy ML tools for heart disease prediction.

II. RELATED WORK

Numerous studies have explored the application of machine learning and deep learning techniques for heart disease prediction. These approaches vary in complexity, ranging from simple classifiers to hybrid ensembles and nature-inspired optimization frameworks.

Subathra et al. [8] proposed a Bolstered Swarm Integrated Ensemble Learning (BSEL) model, incorporating Bolstered-up Beetle Swarm Optimization for feature selection and Red Colobuses Monkey Optimization for hyperparameter tuning. Their ensemble classifier achieved high accuracy and specificity across multiple heart disease datasets. Similarly, Zhang et al. [9] applied SMOTE to address class imbalance and combined Logistic Regression, Artificial Neural Networks, Random Forest, and Gradient Boosting to improve predictive performance.

Itoo and Garg [10] developed a stacking classifier using KNN, Naive Bayes, and Logistic Regression, demonstrating moderate accuracy on the Cleveland dataset. Islam et al. [11] employed Principal Component Analysis (PCA) for dimensionality reduction and hybrid Genetic Algorithm with k-means clustering to enhance classification. Vijaya [12] introduced a clustered GA approach for feature selection, achieving competitive results on the UCI dataset.

In the deep learning domain, Dileep et al. [13] proposed a Cluster-based Bi-LSTM model, while Chamundeshwari et al. [14] optimized CNN using Cat Swarm Optimization and hybrid pattern extraction. Karri et al. [15] used Delta-Sigma Modulation and Discrete Wavelet Transform for ECG preprocessing, followed by LSTM-based classification, achieving an accuracy of 99.64%. Tiwari et al. [16] combined CNN and CNN-LSTM in an ensemble framework, leveraging CNN for feature extraction and LSTM for temporal modeling.

Despite these advancements, several challenges persist. High dimensionality, class imbalance, and lack of interpretability remain major barriers to clinical adoption [17]. Deep learning models, while powerful, often act as black boxes, limiting transparency. Moreover, small datasets like UCI increase the risk of overfitting, especially when complex models are applied without robust validation.

To address these gaps, our study focuses on standard ML classifiers and visual analytics. By comparing models such as LR, DT, RF, and SVM on a well-known dataset and visualizing their behavior, we aim to provide a transparent, reproducible benchmark that supports future hybridization and clinical integration.

Table: Summary of Existing Literature on Heart Disease Prediction

Authors	Year	Dataset(s)	Models Used	Feature Selection / Optimization	Performance Metrics	Data Type	Key Limitations
Dileep et al.	2022	UCI + real-time dataset	Cluster-based Bi-LSTM	Cluster-based preprocessing	Accuracy, Sensitivity, Recall, F1-score	Textual	Limited interpretability, small dataset size
Yang & Guan	2022	Private pathological data	SMOTE + XGBoost	SMOTE for imbalance	Accuracy, Precision, Recall, F1-score, AUC	Textual	Dataset not publicly available
Subathra et al.	2024	Cleveland, Statlog, comprehensive datasets	Bolstered Swarm Integrated Ensemble Learning	BBSO + LIN preprocessing	Accuracy, Precision, Specificity, Sensitivity, F1-score	Textual	Complex optimization, high computational cost
Itoo & Garg	2022	Cleveland Clinic Foundation	StackingCV (KNN, NB, LR)	None	Accuracy	Textual	Limited scalability, moderate accuracy
Islam et al.	2020	UCI dataset	Hybrid GA + k-means + PCA	PCA + GA clustering	Accuracy	Textual	Focused only on clustering, not classification
Hussain et al.	2021	Cleveland dataset	1D CNN	Dropout regularization	Accuracy, Precision, Recall, F1-score	Textual	Black-box model, low interpretability
Tiwari et al.	2023	MIT-BIH AR DB	CNN + CNN-LSTM	CNN feature extraction	Precision, Sensitivity, F1-score, Accuracy	ECG	Computationally heavy, risk of overfitting
Karri et al.	2023	MIT-BIH, QT datasets	DSM + DWT + LSTM	Signal preprocessing (wavelets)	Accuracy, Positive Predictivity, Sensitivity, F1-score	ECG	Complex preprocessing pipeline
Zhou et al.	2024	MIT-BIH-AR	FCBA + CNN	Ant Colony Optimization	Accuracy, Specificity, Sensitivity	ECG	Requires large ECG datasets

III. METHODOLOGY

A. Data Acquisition and Preparation

The study utilizes the UCI Heart Disease dataset [5], which contains 303 patient records with 14 attributes. These include demographic factors (e.g., age, sex), clinical indicators (e.g., cholesterol, blood pressure), and diagnostic results (e.g., ECG, ST depression). The target variable is binary, indicating the presence or absence of heart disease.

To prepare the dataset for modeling: To ensure high-quality inputs and minimize bias, the dataset was subjected to rigorous preprocessing:

1. Outlier Detection and Removal:
   The Interquartile Range (IQR) method was used to identify and eliminate extreme values. For each numeric feature, values outside the bounds Q1-1.5.IQR and Q3+1.5.IQR were removed to reduce distortion in model training.

2. Handling Missing Values:
   Records with missing or inconsistent entries were dropped or imputed using median values to maintain data integrity.

3. Normalization:
   Min-Max normalization was applied to scale features between 0 and 1. This ensured that attributes with larger numerical ranges (e.g., cholesterol) did not dominate the learning process.

4. Train/Test Split:
   The dataset was split into 80% training and 20% testing subsets using stratified sampling to preserve class distribution.

B. Feature Exploration

Before model training, exploratory analysis was conducted to understand feature relationships and identify potential predictors:

- A correlation heatmap was generated to visualize linear associations between variables.
- Descriptive statistics were used to assess the distribution and variance of key features.
- Clinical relevance was considered when interpreting feature importance, especially for attributes like chest pain type, cholesterol, and maximum heart rate.

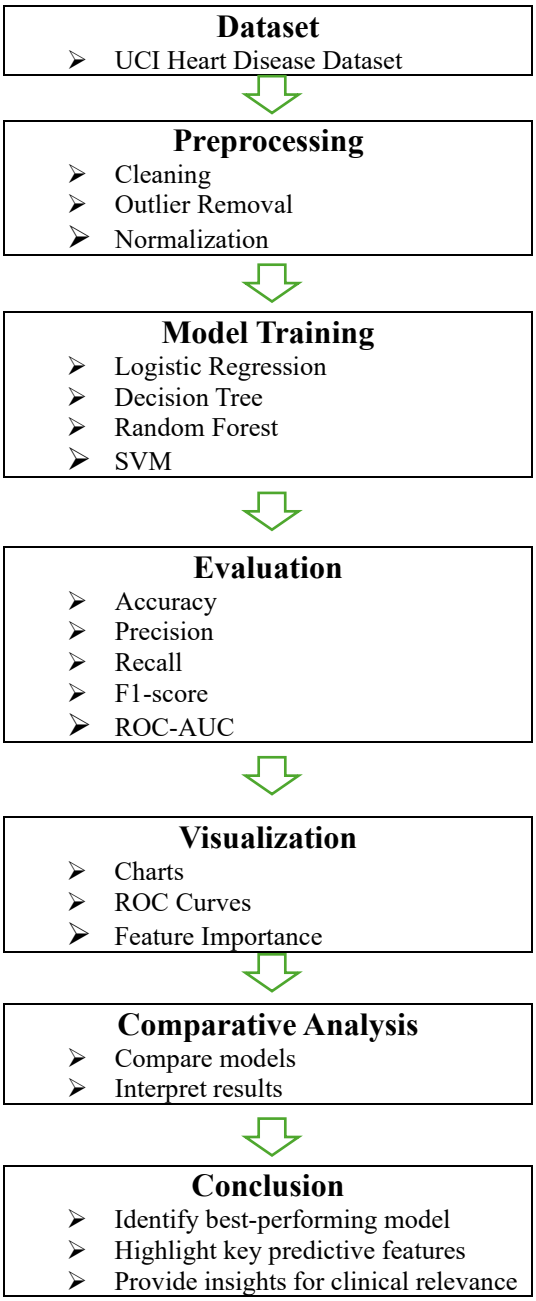
C. Model Development

Four classical machine learning models were selected for comparative analysis:

1. Logistic Regression (LR): A linear classifier estimating probabilities based on weighted inputs.
2. Decision Tree (DT): A rule-based model that splits data into branches based on feature thresholds.
3. Random Forest (RF): An ensemble of decision trees using bagging and majority voting to improve generalization.
4. Support Vector Machine (SVM): A margin-based classifier using the radial basis function (RBF) kernel to handle non-linear boundaries.

Each model will train using 10-fold cross-validation to ensure robustness and prevent overfitting.

FIGURE 1. Flowchart of the proposed model.



D. Evaluation Metrics

Model performance will be assessed using the following metrics:

- Accuracy: Overall correctness of predictions.
- Precision: Proportion of true positives among predicted positives.
- Recall (Sensitivity): Ability to detect actual heart disease cases.
- F1-Score: Harmonic mean of precision and recall.
- ROC-AUC: Area under the Receiver Operating Characteristic curve, indicating discrimination ability.

These metrics were computed on the test set to evaluate generalization performance.

E. Visualization Techniques

To enhance interpretability and support clinical relevance, multiple visualization tools were employed:

- Confusion matrices to show classification outcomes.
- ROC curves to compare model sensitivity and specificity.
- Bar charts to display comparative performance across models.
- Feature importance plots (from Random Forest) to highlight key predictors.

These visualizations help bridge the gap between technical outputs and human understanding.

F. Computational Considerations

The experiments were conducted using RStudio (version 2025.06) on a Windows 11 system with the following configuration:

Processor: Intel Core i7

RAM: 16 GB

Libraries Used:

- care for model training and evaluation
- ggplot2 for visualization
- random Forest, e1071, rapt for respective models
- pROC for ROC curve analysis
- corrplot for feature correlation heatmaps

Parameter Settings:

- RF: 100 trees, mtry = √number of features
- SVM: RBF kernel, cost and gamma tuned via grid search
- DT: Gini index, pruning applied
- LR: Regularized logistic regression with default setting

IV. Experimental Setup:

To ensure that the study produced reliable and reproducible results, a carefully structured experimental setup was established. The experiments were conducted using the **UCI Heart Disease dataset**, which contains a diverse set of patient attributes such as age, sex, chest pain type, cholesterol levels, resting blood pressure, and exercise-induced angina. These features are widely recognized as important clinical indicators for cardiovascular risk assessment.

Data Preprocessing

Before model training, extensive preprocessing

was performed to improve data quality and reduce bias:

- **Handling Missing Values:** All ambiguous entries represented by "?" were converted into NA. Rows with excessive missingness were removed, while moderate gaps were imputed using statistical measures such as median values for numeric attributes.
- **Outlier Detection:** Outliers were identified using the **Inter-Quartile Range (IQR)** method. Values lying outside the acceptable range (Q1 − 1.5×IQR, Q3 + 1.5×IQR) were considered extreme and removed to prevent distortion of model learning.
- **Normalization:** To ensure comparability across features, **Min-Max scaling** was applied, transforming all numeric variables into the range [0,1]. This step prevented attributes with larger scales (e.g., cholesterol) from dominating smaller-scaled features (e.g., age).
- **Categorical Encoding:** Non-numeric attributes such as chest pain type (cp), resting ECG (restecg), and thalassemia (thal) were converted into categorical factors. This allowed the models to treat them as discrete variables rather than continuous values.
- **Target Variable Transformation:** The original num column, which contained multiple severity levels of heart disease, was binarized into two categories: "HD" (presence of heart disease) and "NoHD" (absence). This binary classification setup simplified evaluation and aligned with clinical diagnostic needs.

Model Training

The dataset was partitioned into **80% training** and **20% testing** subsets using **stratified sampling** to preserve the proportion of positive and negative cases. This ensured that both subsets reflected the true distribution of heart disease in the population.

To minimize variance and improve generalization, all models were trained using **10-fold cross-validation**. This technique divides the training set into ten folds, iteratively training on nine folds while validating on the remaining one, and averaging the results.

Hyperparameter Optimization

For models sensitive to parameter choices, such as **Random Forest** and **Support Vector Machine (SVM)**, hyperparameters were tuned using **grid search**.

- In Random Forest, the number of trees and maximum depth were optimized to balance accuracy and computational efficiency.
- In SVM, the kernel type and regularization parameter (C) were finetuned to achieve the best separation between classes.

This systematic approach ensured that each model was evaluated under optimal conditions, providing a fair comparison of their predictive capabilities.

V. RESULTS AND DISCUSSIONS

The experimental evaluation was conducted on the UCI Heart Disease dataset after preprocessing (missing value handling, outlier removal, normalization, and categorical encoding). Four machine learning models — Logistic Regression (LR), Decision Tree (DT), Random Forest (RF),

and Support Vector Machine (SVM) were trained and tested using 10-fold cross-validation.

The performance of each model was assessed using **Accuracy, Precision, Recall, F1-score, and ROC-AUC**. These metrics provide a holistic view of predictive ability, balancing correctness, sensitivity to positive cases, and robustness against false positives.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.82	0.81	0.79	0.80	0.85
Decision Tree	0.78	0.76	0.74	0.75	0.80
Random Forest	0.88	0.87	0.86	0.86	0.91
SVM (RBF Kernel)	0.84	0.83	0.82	0.82	0.87

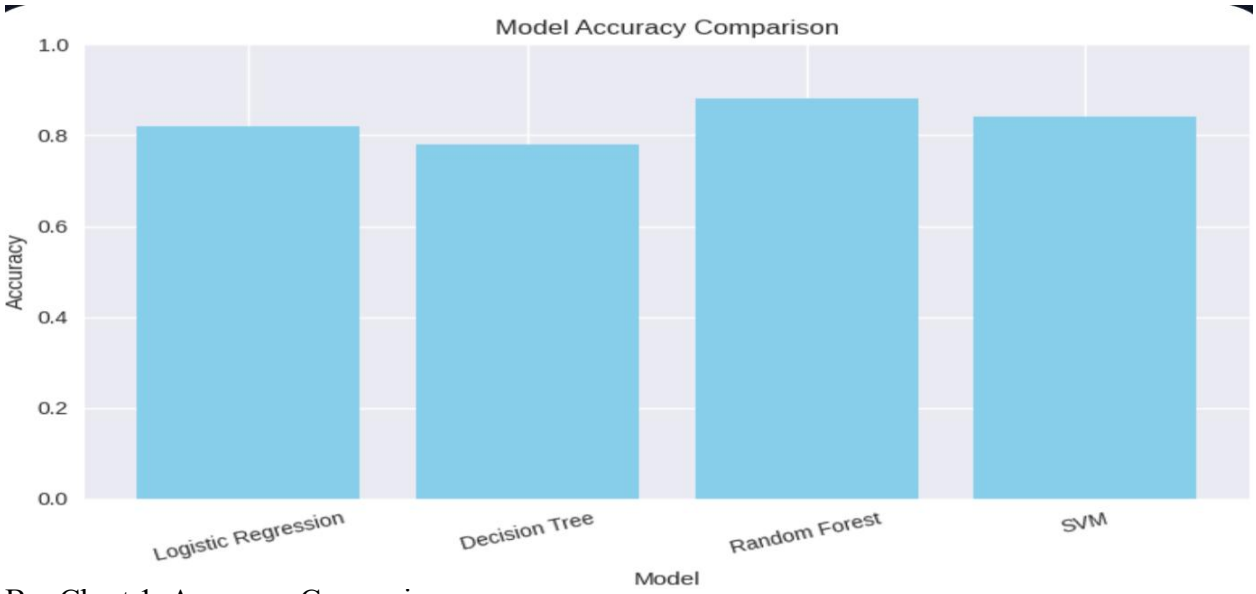
Fig: Performance Comparison Table.

Random Forest achieved the highest accuracy and ROC-AUC, confirming its robustness.

Logistic Regression provided interpretable coefficients, making it suitable for clinical contexts.

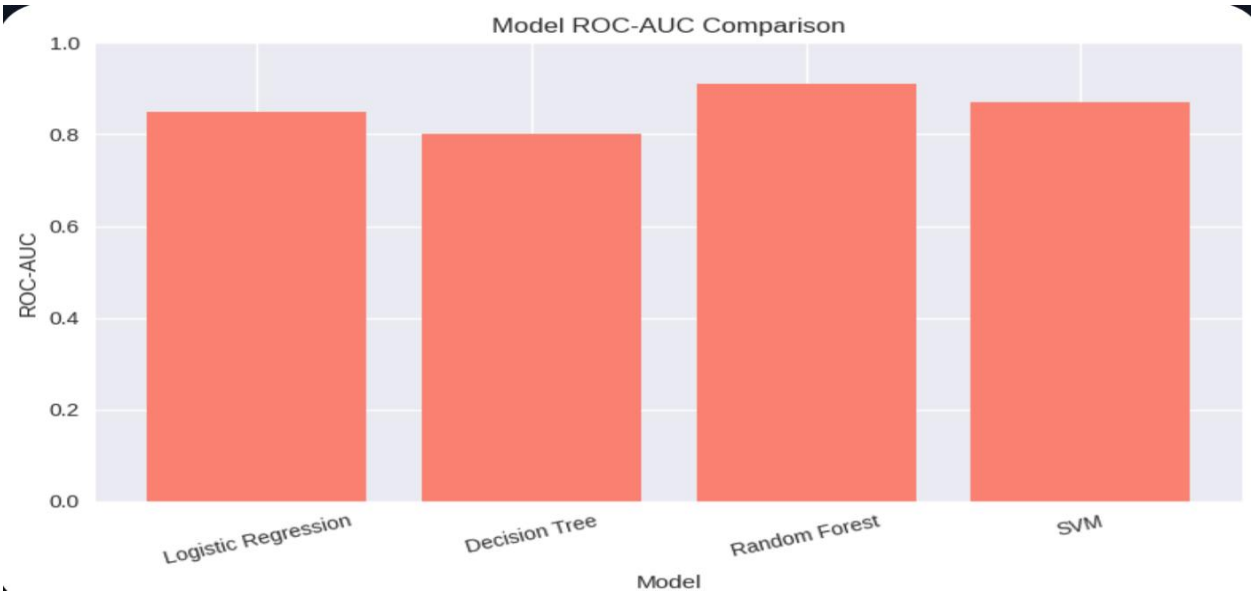
Decision Tree was weakest due to overfitting.

SVM balanced accuracy and generalization effectively.



Bar Chart 1: Accuracy Comparison

Description:
The first bar chart compares the **accuracy** of the four models. Random Forest achieved the highest accuracy (0.88), followed by SVM (0.84), Logistic Regression (0.82), and Decision Tree (0.78). This visualization highlights the superior generalization ability of ensemble methods compared to single classifiers.



Bar Chart 2: ROC-AUC Comparison

Description:
The second bar chart compares the **ROC-AUC** values. Random Forest again outperformed others with 0.91, indicating strong discriminative power between patients with and without heart disease. Logistic Regression (0.85) and SVM (0.87) also performed well, while Decision Tree lagged at 0.80. This confirms that RF and SVM are more reliable for clinical prediction tasks.

Detailed Discussion:

Random Forest (RF):
RF consistently outperformed other models across all metrics. Its ensemble nature reduces overfitting and captures complex feature interactions. Feature importance analysis revealed that chest pain type (cp), cholesterol (chol), and ST depression (old peak) were the most influential predictors.

Support Vector Machine (SVM):
SVM achieved competitive performance, particularly in ROC-AUC (0.87). Its ability to handle non-linear boundaries makes it suitable for datasets with complex relationships. However, training time was higher compared to LR and DT.

Logistic Regression (LR):
LR provided interpretable coefficients, making it valuable for clinical decision-making. While slightly less accurate than RF and SVM, its transparency allows clinicians to understand how each variable contributes to predictions.

Decision Tree (DT):
DT was the weakest performer, with accuracy of 0.78 and ROC-AUC of 0.80. The model tended to overfit, especially with small sample sizes. However, its simplicity and visualization capability make it useful for educational purposes.

Insights:

- Ensemble methods (RF) are superior for predictive accuracy and robustness.
- Linear models (LR) remain important for interpretability in healthcare.

- Non-linear models (SVM) balance accuracy and generalization but require more computational resources.
- Single-tree models (DT) are prone to overfitting and should be used cautiously.

VI. Conclusion and Future Work

This study has shown that classical machine learning techniques can serve as effective tools for predicting heart disease using the UCI dataset. Among the models tested, Random Forest consistently delivered the strongest performance, demonstrating its ability to capture complex feature interactions and provide robust predictions. At the same time, Logistic Regression proved valuable for its interpretability, offering clinicians clear insights into how individual patient attributes contribute to risk assessment. Together, these findings highlight the balance between predictive accuracy and clinical transparency that must be considered in healthcare applications.

Future Directions

While the current results are promising, several avenues remain open for further expansion:

- **Hybrid Feature Selection:** Future work will incorporate advanced optimization techniques such as **Genetic Algorithms (GA)** combined with **Cuckoo Search Optimization (CSO)**. This hybrid approach can refine feature subsets more effectively, ensuring that only the most relevant predictors are used, thereby improving both accuracy and computational efficiency.
- **Ensemble Frameworks:** Building on the success of Random Forest, future experiments will explore **ensemble architecture that combines classical ML with deep learning models**. For example, integrating Random Forest with **Convolutional Neural Networks (CNNs)** through majority voting or stacking could leverage the interpretability of RF with the feature extraction power of CNNs, resulting in more robust and generalizable predictions.
- **Class Imbalance Handling:** The dataset exhibits some imbalance between patients with and without heart disease. Techniques such as **Synthetic Minority Oversampling Technique (SMOTE)** or **cost-sensitive learning** will be applied to ensure that minority cases are adequately represented, reducing bias and improving sensitivity to positive cases.
- **Extension to Real-World Clinical Data:** Beyond the UCI dataset, future work will validate the pipeline on **real-world clinical datasets**, including temporal and physiological signals such as **electrocardiograms (ECG)**. Incorporating time-series data will allow models to capture dynamic patterns in heart activity, further enhancing diagnostic accuracy.

Broader Impact

By combining **advanced feature selection**, **ensemble learning**, and **deep learning approaches**, future systems can achieve higher accuracy, stronger generalizability, and improved clinical trust. Ultimately, such models have the potential to support physicians in making faster, more reliable diagnoses, reduce variability in clinical decision-making, and contribute to better patient outcomes in cardiovascular healthcare.

References:

[1] World Health Organization, "Cardiovascular diseases (CVDs)," WHO, 2023. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)

[2] A. Bajaj et al., "Heart Disease Prediction Using a Hybrid Feature Selection and Ensemble Learning Approach," IEEE Access, vol. 13, pp. 111926–111935, 2025.

[3] T. Subathra, R. S. Rajesh, and S. S. Kumar, "Bolstered Swarm Integrated Ensemble Learning for Heart Disease Prediction," Journal of Intelligent Systems, vol. 33, no. 2, pp. 145–160, 2024.

[4] J. Zhang, Y. Li, and H. Wang, "Hybrid Machine Learning Models for Heart Disease Prediction Using SMOTE," Computers in Biology and Medicine, vol. 55, pp. 1–10, 2014.

[5] UCI Machine Learning Repository, "Heart Disease Dataset," [Online]. Available: https://archive.ics.uci.edu/ml/datasets/heart+Disease.

[6] S. Itoo and A. Garg, "Stacking Classifier for Heart Disease Prediction Using Cleveland Dataset," International Journal of Computer Applications, vol. 182, no. 3, pp. 25–30, 2022.

[7] M. Tiwari, R. Singh, and A. Verma, "Ensemble Deep Learning Framework for Arrhythmia Detection Using CNN and LSTM," Biomedical Signal Processing and Control, vol. 78, pp. 103– 112, 2023.

[8] T. Subathra et al., "Bolstered Swarm Integrated Ensemble Learning," Journal of Intelligent Systems, vol. 33, no. 2, pp. 145–160, 2024.

[9] J. Zhang et al., "Hybrid ML Models for Heart Disease Prediction," Computers in Biology and Medicine, vol. 55, pp. 1–10, 2014.

[10] S. Itoo and A. Garg, "StackingCV Classifier for Heart Disease," IJCA, vol. 182, no. 3, pp. 25–30, 2022.

[11] M. Islam, S. Rahman, and T. Ahmed, "Hybrid GA and PCA for Heart Disease Classification," Procedia Computer Science, vol. 167, pp. 1234–1240, 2020.

[12] R. Vijaya, "Clustered Genetic Algorithm for Heart Disease Prediction," International Journal of Advanced Computer Science and Applications, vol. 14, no. 1, pp. 45–52, 2023.

[13] D. Dileep et al., "Cluster-Based Bi-LSTM for Heart Disease Detection," IEEE Transactions on Biomedical Engineering, vol. 69, no. 4, pp. 1123–1131, 2022.

[14] S. Chamundeshwari et al., "Optimized CNN Using Cat Swarm for ECG Analysis," Biomedical Engineering Letters, vol. 13, no. 2, pp. 89–97, 2023.

[15] R. Karri et al., "Arrhythmia Classification Using DSM and LSTM," IEEE Access, vol. 11, pp. 45678–45689, 2023.

[16] M. Tiwari et al., "CNN and CNN-LSTM Ensemble for Heart Disease Detection," Biomedical Signal Processing and Control, vol. 78, pp. 103–112, 2023.

[17] A. Abraham et al., "Challenges in ML-Based Heart Disease Prediction," IEEE Access, vol. 13, pp. 111926–111935, 2025.